INTRODUCTION

In Class XI you have already learnt about enzymes as biocatalysts. Enzymes are generally considered as essential components for biological reactions.

One of the most captivating areas of study in chemical kinetics is enzyme catalysis. The phenomenon of enzyme catalysis usually results in a very large increase in reaction rate and high specificity. By specificity, we mean that an enzyme is capable of selectively catalysing certain reactants, called substrates.

The specificity of an enzyme for one of the two enantiomers depends on which one fits better than the other into the active site of the enzyme (Figure 9.1).

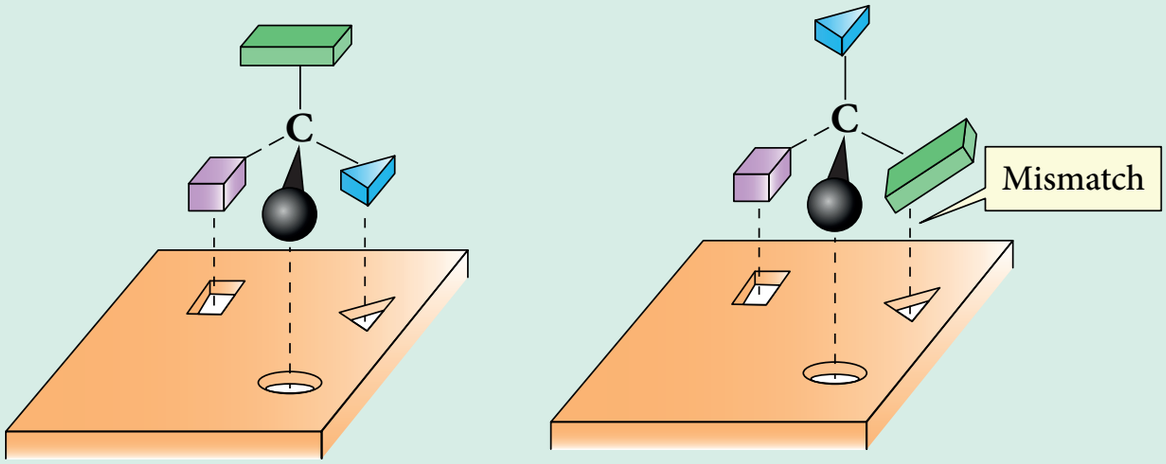


Figure 9.1 Specificity of an enzyme for its substrate

Activation energy is a term coined in 1889 by the Swedish scientist Arrhenius to describe the minimum energy which must be available to the reactants to result in a chemical reaction. An uncatalyzed reaction requires a higher activation energy than does an enzyme-catalyzed reaction (Figure 9.2)

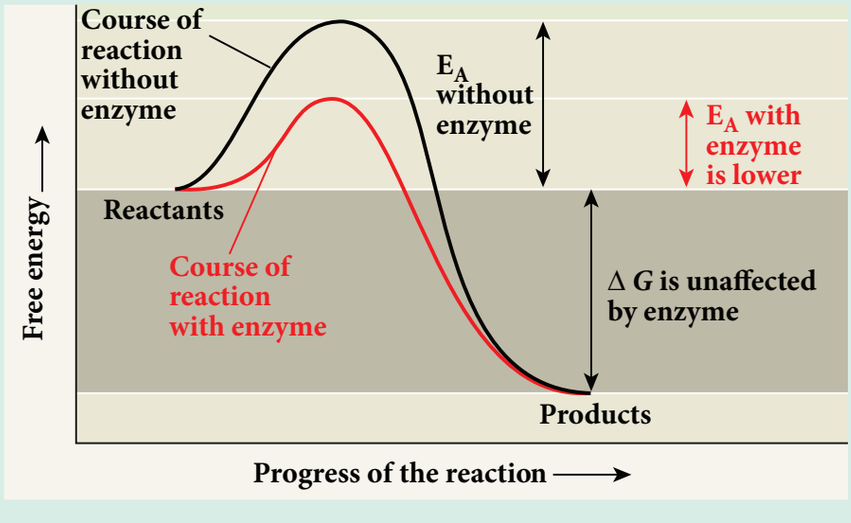


Figure 9.2 Enzymes lower the activation energy barriers.

One important factor which affects the enzyme activity is temperature. All enzymes have the highest activity at a particular temperature which is known as optimum

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